

Azure Service Bus – True and False Rule Filters

1. What are True and False Rule Filters?

Azure Service Bus supports special filters that can be used when we simply want to:

- Accept **all messages**
- Reject **all messages**

These are called:

- **True Rule Filter**
- **False Rule Filter**

True Rule Filter

A True Rule Filter always evaluates to `true`.

Therefore:

The subscription receives all messages sent to the topic.

False Rule Filter

A False Rule Filter always evaluates to `false`.

Therefore:

The subscription does not receive any messages.

2. Simple Example

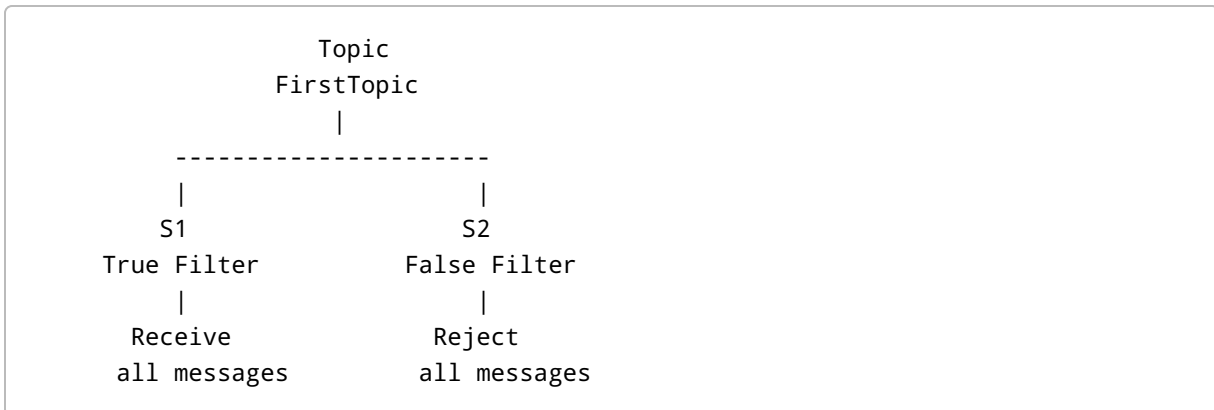
Suppose we have a topic:

FirstTopic

And two subscriptions:

S1 → True Rule Filter
S2 → False Rule Filter

The flow is:



If we send one message:

Test Message

The result will be:

S1 → Message Count = 1
S2 → Message Count = 0

3. True Rule Filter

A **True Rule Filter** always returns `true`.

Conceptually:

1 = 1

Since:

1 = 1

is always true, every message satisfies the filter.

Therefore:

```
Message
↓
True Filter
↓
TRUE
↓
Message delivered to subscription
```

4. False Rule Filter

A **False Rule Filter** always returns `false` .

Conceptually:

```
1 = 0
```

Since:

```
1 = 0
```

is always false, no message satisfies the filter.

Therefore:

```
Message
↓
False Filter
↓
FALSE
↓
Message NOT delivered to subscription
```

5. Why Do We Use True and False Filters?

These filters are useful when we want to control whether a subscription receives messages.

True Filter

Use when:

We want the subscription to receive all messages.

False Filter

Use when:

We temporarily don't want the subscription to receive any messages.

They are also useful when creating a subscription first and planning to add or modify filtering rules later.

6. Creating Subscription S1 with True Filter

First, create the subscription options:

```
var s1Details =  
    new CreateSubscriptionOptions(  
        topicName,  
        "S1");
```

Here:

topicName → Topic name

S1 → Subscription name

7. Create the True Rule Filter

Create a True Rule Filter:

```
var trueRule =  
    new CreateRuleOptions(  
        "TrueRule",  
        new SqlRuleFilter("1=1"));
```

The important part is:

```
1 = 1
```

This condition is always true.

Therefore, every message matches this filter.

8. Create Subscription S1

Now create the subscription:

```
await adminClient.CreateSubscriptionAsync(  
    s1Details,  
    trueRule);
```

The resulting structure is:

```
Topic  
├── S1  
│   ├── TrueRule  
│   │   ├── SQL Filter  
│   │   │   └── 1 = 1
```

9. Creating Subscription S2 with False Filter

Now create another subscription:

```
var s2Details =  
    new CreateSubscriptionOptions(  
        topicName,  
        "S2");
```

Then create the False Rule Filter:

```
var falseRule =  
    new CreateRuleOptions(  
        "FalseRule",  
        new SqlRuleFilter("1=0"));
```

The condition:

1 = 0

is always false.

10. Create Subscription S2

Create the subscription:

```
await adminClient.CreateSubscriptionAsync(  
    s2Details,  
    falseRule);
```

The resulting structure is:

```
Topic  
├── S2  
│   ├── FalseRule  
│   │   └── SQL Filter  
│   │       1 = 0
```

11. Complete Example

The complete concept can be written as:

```
var adminClient =  
    new ServiceBusAdministrationClient(connectionString);
```

```
// S1 - True Filter
var s1Details =
    new CreateSubscriptionOptions(
        topicName,
        "S1");

var trueRule =
    new CreateRuleOptions(
        "TrueRule",
        new SqlRuleFilter("1=1"));

await adminClient.CreateSubscriptionAsync(
    s1Details,
    trueRule);

// S2 - False Filter
var s2Details =
    new CreateSubscriptionOptions(
        topicName,
        "S2");

var falseRule =
    new CreateRuleOptions(
        "FalseRule",
        new SqlRuleFilter("1=0"));

await adminClient.CreateSubscriptionAsync(
    s2Details,
    falseRule);
```

12. Verify in Azure Portal

After executing the application:

```
Azure Portal
↓
Service Bus Namespace
↓
Topic
↓
Subscriptions
```

You should see:

S1
S2

13. Verify S1

Open subscription `S1`.

You should see the rule:

TrueRule

with a filter similar to:

1 = 1

This means:

1 = 1
↓
TRUE

Therefore:

S1 accepts all messages.

14. Verify S2

Open subscription `S2`.

You should see:

FalseRule

with a filter:

1 = 0

This means:

1 = 0
↓
FALSE

Therefore:

S2 rejects all messages.

15. Send a Test Message

Now send a message to the topic using Service Bus Explorer.

Example:

Message Body:

Test Message

The message is sent to:

FirstTopic

The topic then evaluates the subscriptions.

16. How S1 Processes the Message

S1 has:

1 = 1

The filter evaluates:

```
1 = 1
↓
TRUE
```

Therefore:

```
Test Message
  ↓
  S1
  ↓
Received ✓
```

S1 message count becomes:

```
1
```

17. How S2 Processes the Message

S2 has:

```
1 = 0
```

The filter evaluates:

```
1 = 0
↓
FALSE
```

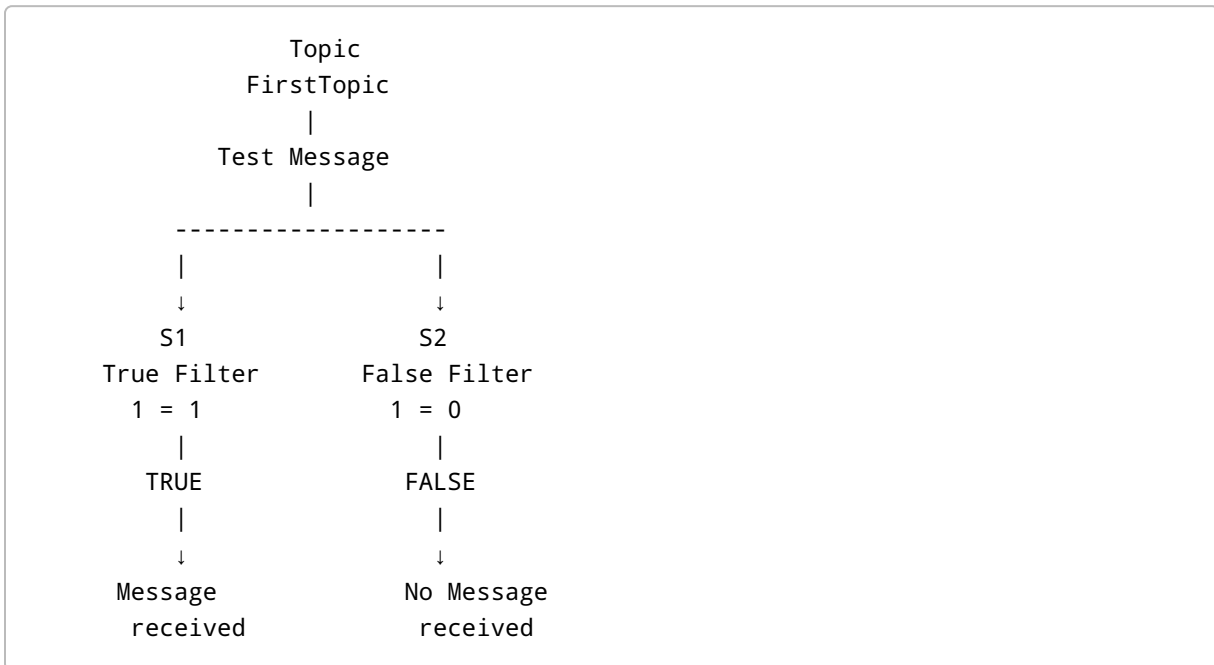
Therefore:

```
Test Message
  ↓
  S2
  ↓
Not Received ✗
```

S2 message count remains:

0

18. Complete Flow



19. Important Difference

Filter	Condition	Result
True Rule Filter	1 = 1	Always TRUE
False Rule Filter	1 = 0	Always FALSE

True Filter

1 = 1

Means:

Accept every message.

False Filter

```
1 = 0
```

Means:

Reject every message.

20. Real-Time Use Case

Suppose we have an application with:

```
OrderTopic
```

and a subscription:

```
OrderProcessing
```

Initially, we don't want the subscription to process any messages.

We can create it with:

```
False Filter
```

```
1 = 0
```

Now:

```
Messages → OrderTopic
           ↓
        OrderProcessing
           ↓
         1 = 0
           ↓
        No messages
```

Later, if we change the filtering rule to a True Filter:

1 = 1

then:

```
Messages → OrderTopic
          ↓
        OrderProcessing
          ↓
         1 = 1
          ↓
      All messages
```

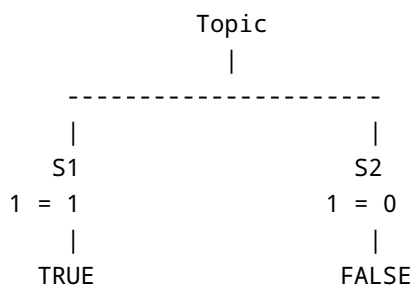
21. Important Point About Topic and Subscription

Remember that a topic does not directly send messages to one specific subscription.

The basic flow is:

```
Producer
  ↓
Topic
  ↓
Subscriptions
  ↓
Filters decide
  ↓
Matching subscriptions receive messages
```

For example:



|
Receive

|
Reject

22. True Filter vs Correlation Filter vs SQL Filter

There are different filtering approaches.

True Rule Filter

`1 = 1`

Accepts every message.

False Rule Filter

`1 = 0`

Rejects every message.

Correlation Filter

Example:

`City = Mumbai`

Used for simple property-based equality matching.

SQL Filter

Example:

`City = 'Mumbai' AND OrderAmount > 10000`

Used for SQL-like and more complex filtering conditions.

23. Quick Comparison

Filter Type	Example	Purpose
True Filter	<code>1 = 1</code>	Accept all messages
False Filter	<code>1 = 0</code>	Reject all messages
Correlation Filter	<code>City = Mumbai</code>	Simple property matching
SQL Filter	<code>City = 'Mumbai' AND Amount > 10000</code>	Complex filtering

24. Interview Questions

Q1. What is a True Rule Filter?

A True Rule Filter is a filter that always evaluates to `true`. Therefore, the subscription receives all messages.

Example:

```
1 = 1
```

Q2. What is a False Rule Filter?

A False Rule Filter always evaluates to `false`. Therefore, the subscription does not receive any messages.

Example:

```
1 = 0
```

Q3. What is the difference between True and False Rule Filters?

```
True Filter:
1 = 1
→ Always TRUE
→ Receives all messages
```

False Filter:
1 = 0
→ Always FALSE
→ Receives no messages

Q4. Does a False Filter delete the message?

No.

A false filter simply means that the particular subscription does not receive the message.

Other subscriptions may still receive the same message if their filters match.

Q5. Can another subscription receive a message when S2 has a False Filter?

Yes.

For example:

```
Topic
|
|--- S1 → 1 = 1 → Receives ✓
|
|--- S2 → 1 = 0 → Does not receive ✗
```

S1 can receive the message even though S2 rejects it.

25. Key Points for Quick Revision

- **True Rule Filter** always evaluates to `true`.
- Example:

1 = 1

- A subscription with a True Filter receives **all messages**.
- **False Rule Filter** always evaluates to `false`.

- Example:

1 = 0

- A subscription with a False Filter receives **no messages**.
- Filtering happens at the **subscription level**.
- One subscription can receive a message while another subscription does not.
- A False Filter does not mean the message is deleted.
- True and False filters are useful for controlling message delivery.

26. One-Line Interview Answer

A True Rule Filter always evaluates to true and allows all messages to a subscription, while a False Rule Filter always evaluates to false and prevents all messages from being delivered to that subscription.

Easy Memory Trick

```
1 = 1
  ↓
TRUE
  ↓
Receive ALL

1 = 0
  ↓
FALSE
  ↓
Receive NONE
```